Engine Model(s): PT6A

72-30-05

COMPRESSOR ROTOR - DESCRIPTION AND OPERATION

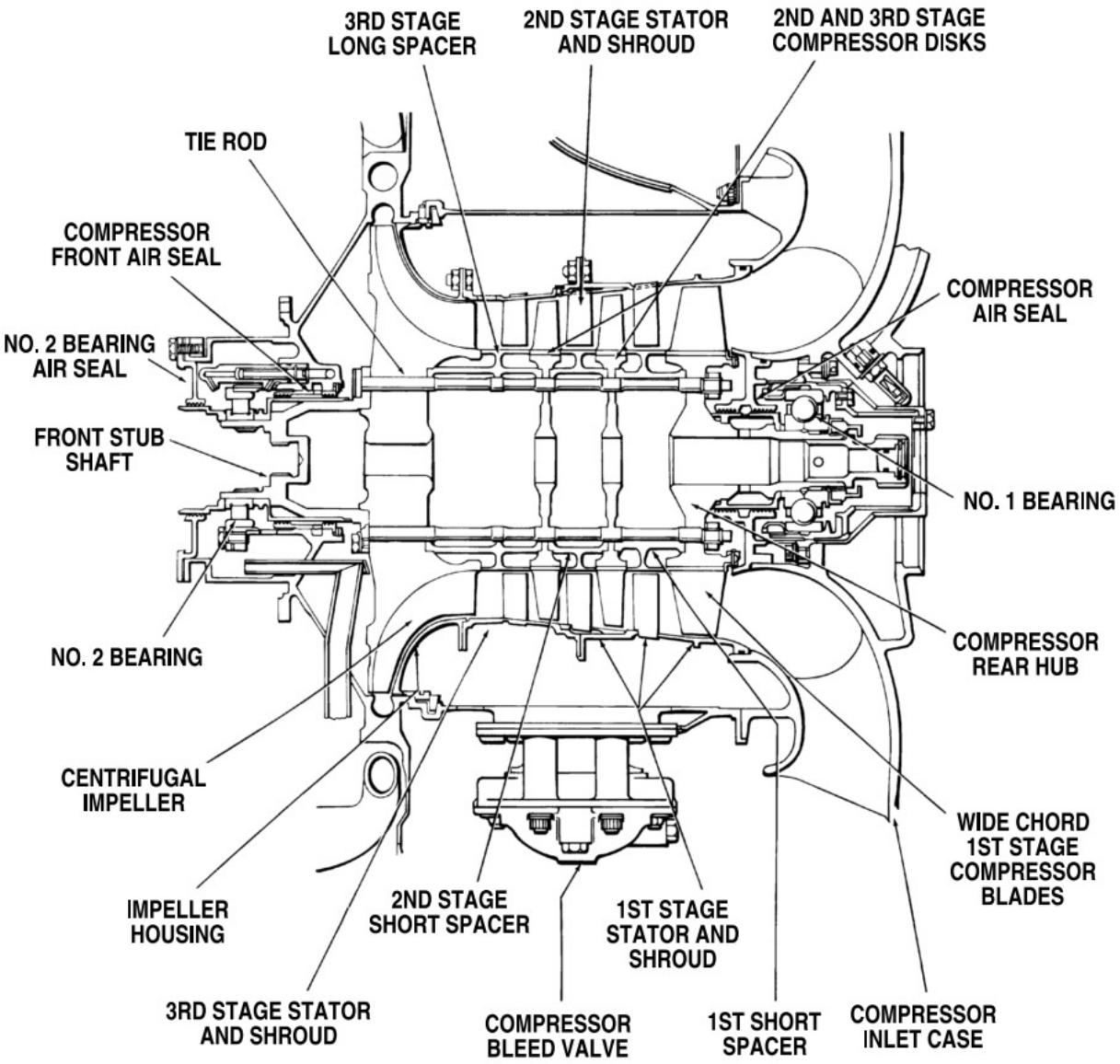
1. General

The compressor rotor is housed in the rear section of the gas generator case and is driven by the compressor turbine. It compresses incoming air for delivery to the combustion chamber.

2. Description and Operation (Ref. Fig. 1)

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Figure 1 Compressor Rotor - Cross-section



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The compressor rotor comprises a three axial stages, three interstage spacers and a single-stage centrifugal impeller. First-stage rotor blades are of titanium, while second- and third-stage rotor blades are cadmium-plated stainless steel. The rotor blades fit into dovetail grooves machined in the rims of respective disks. The limited clearance between the blade roots and grooves produces the characteristic clicking heard during compressor run-down. Axial movement of the blades is limited by interstage spacers installed between the rotor disks. The airfoil section of the first-stage blades differs from that of the second- and third-stage blades. The first-stage rotor has 16 blades while the second- and third-stage rotors each have 32 blades. The centrifugal impeller is bolted to the three axial stages and associated spacers by six numbered tie rods to form an integral unit.

The first-stage and second-stage stators have 44 vanes; the third-stage has 40 vanes. The vanes in each stage are of stainless steel. Each set of vanes is held in position by its respective compressor stator assembly through which the outer ends of the vanes protrude and to which the vane ends are brazed. The first- and third-stage stator assemblies are bolted together, while the second-stage stator assembly is retained between flanges in the first- and third-stage stator assemblies. The third-stage stator is also bolted to the impeller housing. Rotation is prevented by slots and lugs in the mating flanges. Interstage pressure (P2.5) is vented to the compressor bleed valve chamber in the gas generator case through slots in the third-stage stator assembly.

The compressor stubshaft, centrifugal impeller, impeller housing and third-stage compressor stator assembly are positioned, in that order from front to rear, followed by the third-stage disk spacer and rotor disk, the second-stage stator assembly, second-stage disk spacer and rotor disk, the first-stage stator assembly, first-stage disk spacer and compressor rear hub (first-stage disk and blade assembly). The impeller housing is secured to the gas generator case by a wedge shaped retaining ring while the first-stage stator assembly is retained within a flange on the front face of the compressor inlet case. Sealing is accomplished by a preformed packing at the interface.

The compressor front stubshaft is a hollow steel forging, machined to accommodate the compressor air seal rotor and the No. 2 bearing assembly. The No. 2 bearing is a roller-type and supports the stubshaft and forward end of the compressor rotor assembly in the gas generator case. The compressor turbine rotor seal is fitted on the stubshaft forward of the No. 2 bearing.

The compressor rear hub is an integral part of the first-stage compressor rotor disk. The hub consists of a hollow component machined to accommodate the No. 1 bearing rotor air seal, adjusting spacer, and the No. 1 bearing. The No. 1 bearing is a ball-type and supports the compressor rear hub and the rear end of the compressor assembly in the compressor inlet case. A short, hollow, steel coupling, externally splined at the rear end, is secured in the hollow end of the rear hub by a transverse pin which passes through the hub and coupling.

A hollow steel coupling shaft, internally splined at each end, extends the compressor drive rearward to drive the accessory gearbox. The front end is secured by a ball-lock arrangement while the rear end is splined to the gearbox drive shaft.

The No. 1 and 2 bearings support the compressor assembly in the rear section of the gas generator case. The No. 1 bearing outer race is held in the adapter of the flexible housing and adapter assembly by a keywasher and spanner nut. The outer housing and stator air seal are bolted to the inlet case. The split inner race of the No. 1 bearing, spacer and rotor air seal are stacked against a shoulder on the rear hub and secured by a keywasher and spanner nut. A puller groove embodied in

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the OD of the rear half of the inner race facilitates removal.

The flange of the No. 2 bearing outer race is secured to the centerbore of the gas generator case by four bolts and tablock washers. The inner race of the No. 2 bearing is interposed between compressor rotor air seal and the compressor turbine rotor air seal, with the compressor rotor air seal bearing against a shoulder on the stubshaft. The bearing and rotor air seals are retained by the compressor turbine disk.

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